

OpenStax Calculus Volume 3

Section 4.5 Exercises: The Chain Rule

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For the following exercises, use the information provided to solve the problem.

1. Let $w(x, y, z) = xy \cos z$, where $x = t$, $y = t^2$, and $z = \arcsin t$. Find $\frac{dw}{dt}$.
2. Let $w(t, v) = e^{tv}$ where $t = r + s$ and $v = rs$. Find $\frac{\partial w}{\partial r}$ and $\frac{\partial w}{\partial s}$.
3. If $w = 5x^2 + 2y^2$, $x = -3s + t$, and $y = s - 4t$, find $\frac{\partial w}{\partial s}$ and $\frac{\partial w}{\partial t}$.
4. If $w = xy^2$, $x = 5 \cos(2t)$, and $y = 5 \sin(2t)$, find $\frac{dw}{dt}$.
5. If $f(x, y) = xy$, $x = r \cos \theta$, and $y = r \sin \theta$, find $\frac{\partial f}{\partial r}$ and express the answer in terms of r and θ .
6. Suppose $f(x, y) = x + y$, where $x = r \cos \theta$ and $y = r \sin \theta$. Find $\frac{\partial f}{\partial \theta}$.

For the following exercises, find $\frac{df}{dt}$ using the chain rule and direct substitution.

7. $f(x, y) = x^2 + y^2$, $x = t$, $y = t^2$
8. $f(x, y) = \sqrt{x^2 + y^2}$, $y = t^2$, $x = t$
9. $f(x, y) = xy$, $x = 1 - \sqrt{t}$, $y = 1 + \sqrt{t}$
10. $f(x, y) = \frac{x}{y}$, $x = e^t$, $y = 2e^t$
11. $f(x, y) = \ln(x + y)$, $x = e^t$, $y = e^t$
12. $f(x, y) = x^4$, $x = t$, $y = t$
13. Let $w(x, y, z) = x^2 + y^2 + z^2$, $x = \cos t$, $y = \sin t$, and $z = e^t$. Express w as a function of t and find $\frac{dw}{dt}$ directly. Then, find $\frac{dw}{dt}$ using the chain rule.
14. Let $z = x^2y$, where $x = t^2$ and $y = t^3$. Find $\frac{dz}{dt}$.
15. Let $u = e^x \sin y$, where $x = -\ln(2t)$ and $y = \pi t$. Find $\frac{du}{dt}$ when $x = \ln 2$ and $y = \frac{\pi}{4}$.

For the following exercises, find $\frac{dy}{dx}$ using partial derivatives.

16. $\sin(6x) + \tan(8y) + 5 = 0$
17. $x^3 + y^2x - 3 = 0$

18. $\sin(x + y) + \cos(x - y) = 4$
19. $x^2 - 2xy + y^4 = 4$
20. $xe^y + ye^x - 2x^2y = 0$
21. $x^{2/3} + y^{2/3} = a^{2/3}$
22. $x \cos(xy) + y \cos x = 2$
23. $e^{xy} + ye^y = 1$
24. $x^2y^3 + \cos y = 0$
25. Find $\frac{dz}{dt}$ using the chain rule where $z = 3x^2y^3$, $x = t^4$, and $y = t^2$.
26. Let $z = 3 \cos x - \sin(xy)$, $x = \frac{1}{t}$, and $y = 3t$. Find $\frac{dz}{dt}$.
27. Let $z = e^{1-xy}$, $x = t^{1/3}$, and $y = t^3$. Find $\frac{dz}{dt}$.
28. Find $\frac{dz}{dt}$ by the chain rule where $z = \cosh^2(xy)$, $x = \frac{1}{2}t$, and $y = e^t$.
29. Let $z = \frac{x}{y}$, $x = 2 \cos u$, and $y = 3 \sin v$. Find $\frac{\partial z}{\partial u}$ and $\frac{\partial z}{\partial v}$.
30. Let $z = e^{x^2y}$, where $x = \sqrt{uv}$ and $y = \frac{1}{v}$. Find $\frac{\partial z}{\partial u}$ and $\frac{\partial z}{\partial v}$.
31. If $z = xye^{x/y}$, $x = r \cos \theta$, and $y = r \sin \theta$, find $\frac{\partial z}{\partial r}$ and $\frac{\partial z}{\partial \theta}$ when $r = 2$ and $\theta = \frac{\pi}{6}$.
32. Find $\frac{\partial w}{\partial s}$ if $w = 4x + y^2 + z^3$, $x = e^{rs^2}$, $y = \ln(\frac{r+s}{t})$, and $z = rst^2$.
33. If $w = \sin(xyz)$, $x = 1 - 3t$, $y = e^{1-t}$, and $z = 4t$, find $\frac{\partial w}{\partial t}$.

For the following exercises, use this information: A function $f(x, y)$ is said to be homogeneous of degree n if $f(tx, ty) = t^n f(x, y)$. For all homogeneous functions of degree n , the following equation is true: $x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} = n f(x, y)$. Show that the given function is homogeneous and verify that $x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} = n f(x, y)$.

34. $f(x, y) = 3x^2 + y^2$
35. $f(x, y) = \sqrt{x^2 + y^2}$
36. $f(x, y) = x^2y - 2y^3$
37. The volume of a right circular cylinder is given by $V(x, y) = \pi x^2y$, where x is the radius of the cylinder and y is the cylinder height. Suppose x and y are functions of t given by $x = \frac{1}{2}t$ and $y = \frac{1}{3}t$ so that x and y are both increasing with time. How fast is the volume increasing when $x = 2$ and $y = \frac{4}{3}$?
38. The pressure P of a gas is related to the volume and temperature by the formula $PV = kT$, where temperature is expressed in kelvins. Express the pressure of the gas as a function of both V and T . Find $\frac{dP}{dt}$ when $k = 1$, $\frac{dV}{dt} = 2 \text{ cm}^3/\text{min}$, $\frac{dT}{dt} = \frac{1}{2} \text{ K/min}$, $V = 20 \text{ cm}^3$, and $T = 20^\circ\text{F}$.
39. The radius of a right circular cone is increasing at 3 cm/min whereas the height of the cone is decreasing at 2 cm/min . Find the rate of change of the volume of the cone when the radius is 13 cm and the height is 18 cm .

40. The volume of a frustum of a cone is given by the formula $V = \frac{1}{3}\pi z(x^2 + y^2 + xy)$, where x is the radius of the smaller circle, y is the radius of the larger circle, and z is the height of the frustum (see figure). Find the rate of change of the volume of this frustum when $x = 10$ in., $y = 12$ in., and $z = 18$ in. if $\frac{dz}{dt} = -5$, $\frac{dx}{dt} = 1$, $\frac{dy}{dt} = 1$ (all in/min).
[Figure: A conical frustum (that is, a cone with the pointy end cut off) with height z , larger radius y , and smaller radius x .]
41. A closed box is in the shape of a rectangular solid with dimensions x , y , and z . (Dimensions are in inches.) Suppose each dimension is changing at the rate of 0.5 in./min. Find the rate of change of the total surface area of the box when $x = 2$ in., $y = 3$ in., and $z = 1$ in.
42. The total resistance in a circuit that has three individual resistances represented by x , y , and z is given by the formula $R(x, y, z) = \frac{xyz}{yz+xz+xy}$. Suppose at a given time the x resistance is 100Ω , the y resistance is 200Ω , and the z resistance is 300Ω . Also, suppose the x resistance is changing at a rate of $2\Omega/\text{min}$, the y resistance is changing at the rate of $1\Omega/\text{min}$, and the z resistance has no change. Find the rate of change of the total resistance in this circuit at this time.
43. The temperature T at a point (x, y) is $T(x, y)$ and is measured using the Celsius scale. A fly crawls so that its position after t seconds is given by $x = \sqrt{1+t}$ and $y = 2 + \frac{1}{3}t$, where x and y are measured in centimeters. The temperature function satisfies $T_x(2, 3) = 4$ and $T_y(2, 3) = 3$. How fast is the temperature increasing on the fly's path after 3 sec?
44. The x and y components of a fluid moving in two dimensions are given by the following functions: $u(x, y) = 2y$ and $v(x, y) = -2x$; $x \geq 0$, $y \geq 0$. The speed of the fluid at the point (x, y) is $s(x, y) = \sqrt{u(x, y)^2 + v(x, y)^2}$. Find $\frac{\partial s}{\partial x}$ and $\frac{\partial s}{\partial y}$ using the chain rule.
45. Let $u = u(x, y, z)$, where $x = x(w, t)$, $y = y(w, t)$, $z = z(w, t)$, $w = w(r, s)$, and $t = t(r, s)$. Use a tree diagram and the chain rule to find an expression for $\frac{\partial u}{\partial r}$.